

JD Cortex — Extended Work Proposal

Selective memory erasure, Urdu conversational support,
and autonomous movement

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JD Cortex is a working perception, reasoning, and memory system for an EZ-Robot JD Humanoid. The robot recognises enrolled people, records a timestamped episodic diary of who arrived, left, and what they were doing, answers questions about what it has seen, and executes only actions validated against a hand-tested whitelist. It has been demonstrated end to end on hardware.

This document proposes four extensions, in the order we would take them. Each states what exists today, what we propose and its impact.

1 Urdu Conversational Support

Speech-to-text runs locally through a Parakeet model, which is English-only. The language model handling conversation is already multilingual, but nothing in the pipeline captures Urdu speech, and replies are spoken through ARC’s English voice.

Proposal: An Urdu path through the same pipeline rather than a parallel system: an Urdu-capable speech-to-text engine selectable alongside the existing one; a language flag threaded through the prompt so the spoken reply comes back in Urdu; and Urdu replies routed to a separate text-to-speech engine, leaving the English path untouched.

One detail matters for correctness. The model’s reply carries two lines: the sentence to speak, and a matched action name. The action name is compared literally against the whitelist, so it must stay in its existing English form even when the conversation is in Urdu. The same applies to the internal scene and diary text the model reads — only user-facing speech changes language.

Impact: Real commands will code-switch (“JD, aagay chalo”), and Urdu is comparatively low-resource for speech recognition, with code-switched Urdu-English harder still. Model choice therefore has to be validated against recorded commands rather than assumed. Whisper (`large-v3`, run locally through `faster-whisper`) is the practical starting point since it is multilingual without fine-tuning; AI4Bharat’s IndicConformer is the comparison worth running, as models trained on South Asian languages may handle local accents better.

First pass: Add Whisper as a second recognition path selectable from the panel; add the language flag; route Urdu replies through a local Urdu voice; and evaluate on roughly twenty recorded commands spanning movement, diary questions, and casual conversation, in pure Urdu and code-switched.

2 Autonomous Movement and Physical Safety

All spatial awareness comes from a single camera, which gives direction but not range: the robot can see a cup and not know how far away it is, and has no way to tell that the floor ends. Movement is limited to fixed, pre-approved actions triggered by a command or a gesture — the robot never decides how far to walk.

Proposal:

- A forward-facing distance sensor (ultrasonic or time-of-flight) to supply range.
- Downward infrared reflectance sensors at the feet to detect a step edge, stairs, or a table edge from the sudden loss of a reflected signal.
- **An automatic ledge stop** that interrupts any walk in progress, whatever started it — voice, gesture, or autonomous approach. This mirrors a safety pattern already in the system: gesture control halts the robot the moment the controlling hand leaves frame.
- **Walk-to-object.** Face the target using its horizontal position in frame, take a bounded number of steps, re-check camera and range between steps, and stop when range confirms arrival. Deliberately iterative, because computing a distance once and walking blind fails as soon as the target or the robot’s footing shifts.

This complements the camera rather than replacing it. Vision answers *what* and roughly *where*; the sensors answer *how far* and *is the ground safe*, which one camera cannot reliably do.

Sequencing: The ledge stop is built and proven before any autonomous approach behaviour. It protects every future movement feature rather than just this one, and a legged robot walking off a table edge on a valid command is a real and irreversible hardware loss.

3 Selective Memory Erasure

The diary is a structured SQLite store, indexed by person and time, read fresh into the language model’s prompt on every question. There is no vector index, no fine-tuned model, and no cached summary: the record the robot answers from is the record on disk. Enrolled faces are stored separately as numeric encodings, and captured snapshots as image files.

Proposal: A per-person erasure operation — requestable from the ARC panel, or spoken by the person themselves — that removes an individual’s episodic history and their biometric enrolment together, and can be shown to have done so.

Impact: Removing one person’s data from a trained model is an open research problem, precisely because the information is diffused through the weights. JD Cortex is the opposite case by construction. Because memory is externalised, structured, and re-read per query rather than learned, erasure can be exact rather than approximate — which makes the system a rare testbed for questions that are difficult to study in end-to-end models:

- **Deletion versus de-identification.** Deleting a person’s arrival and departure events changes the historical record: durations, counts, and “who was here” answers all shift. Folding them into the existing anonymous *unknown* bucket instead preserves that somebody was present while removing who it was. Which is correct depends on whose interest is being served, and both are implementable here.
- **Residual leakage.** Someone whose own rows are gone can still be re-identified through co-occurrence, aggregate counts, or a snapshot taken at the time. Snapshots currently carry no per-person association, so honest erasure requires a schema addition, not just a `DELETE`.
- **Verifiable erasure.** An audit record that a deletion occurred and what its scope was, while retaining none of the content deleted.

A robot that logs identified people should be able to forget one of them on request. Our architecture is unusual in that it can actually prove it did.

First pass. Erase by name across diary rows and enrolled encodings; add per-snapshot person association so images are covered; write the audit entry; expose the operation as a panel action and a spoken request. The whole feature is demonstrable in one sitting: enrol, be seen, ask the robot what it remembers, request erasure, ask again.

4 Scaling Beyond One Robot

One robot, one ARC instance, one machine running every process side by side, coordinated through local files and a single SQLite database. Nothing in the code has any concept of *which* robot.

Proposal: Add a robot identifier to the diary schema and scene state while still running a single unit, so the schema is ready before it is needed. Replace file polling with a lightweight message bus on that single robot, to validate the approach before it has to coordinate anything. Separate frame capture from inference, so the heavy vision computation can run on a GPU machine that is not the one talking to the robot. Only then attempt a second concurrent unit against a shared diary.

The third of these is worth doing on its own merits even if multi-robot never happens: it is what lets the vision pipeline run at full model size without the robot-facing process competing for the same hardware.

5 Resources Requested

Resource	What it unblocks
Paid language-model API tier	The most limiting constraint today. The free tier allows roughly twenty requests per day per model, which caps a full test session at a handful of conversational turns and has been the main brake on evaluation. Low cost, immediate effect.
GPU access (lab workstation or a server allocation)	Whisper <code>large-v3</code> and full-size detection models at usable latency; also the prerequisite for separating vision compute from robot control.
Distance and ledge sensors, plus a microcontroller bridge	Ultrasonic or time-of-flight sensor, a pair of infrared reflectance sensors, and an Arduino/ESP32 to bridge them back. Hobby-grade parts; the enabling hardware for every autonomous movement feature.
Recorded Urdu command set	Participant time to record roughly twenty spoken commands, pure Urdu and code-switched, as the evaluation set for speech-recognition model selection.
Access to a second JD unit	Multi-robot validation. Needed only at the final stage, and only if the earlier scaling steps succeed.

6 Proposed Sequence

Phase	Work	Demonstrable outcome
First	Selective erasure, first pass. Urdu recognition and speech path.	The robot converses in Urdu, and forgets a named person on request, verifiably.
Second	Ledge-stop safety and distance sensing. Snapshot association and erasure audit.	The robot refuses to walk off an edge, under any command. Erasure covers images as well as records.
Third	Walk-to-object. Robot identifier in the schema, message bus, separated vision compute.	The robot approaches a named object under its own judgement. Architecture ready for a second unit.

None of this requires abandoning the current architecture. The project's core design — separate processes for sensing, remembering, deciding, and acting, coordinated through simple shared state — extends naturally in each of these directions. The erasure work is the most novel and the best candidate for publication; the Urdu work is the most self-contained and the easiest to demonstrate; the sensor work is the most safety-critical, and is the first feature that would let JD move under its own judgement rather than only on direct command.